Project Design Phase-I
Solution Architecture

Date	06 March 2026
Team ID	699d5f83fdf409fdbd1bec82
Project Name	InsureGuard — AI Insurance Fraud Detection System
Maximum Marks	4 Marks

Solution Architecture:

For the InsureGuard project, the solution architecture serves as the foundational blueprint that connects the business problem of insurance fraud with our automated machine learning system. Its primary goals are to:

- Map the business requirement (reducing financial losses from fraud) directly to a technical workflow (an automated ML classification pipeline).
- Illustrate how claim data flows from the user interface through data preprocessing, model inference, and final risk scoring.
- Define the specific technologies (Streamlit, Scikit-learn, Python) handling each phase of the application.
- Provide a clear, transparent framework for stakeholders to understand how the system is deployed, managed, and scaled.

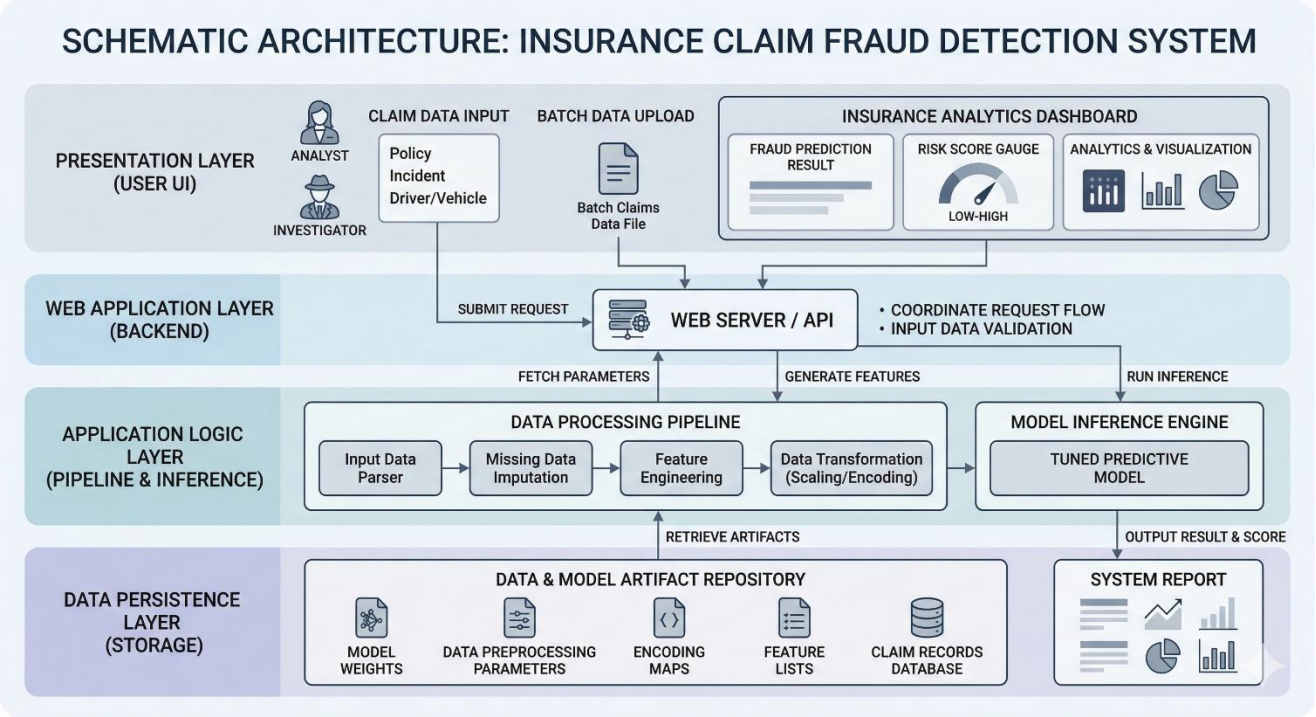

Figure 1: Architecture and data flow of the InsureGuard auto insurance fraud detection system